2022

EXAMINATION SERIES FOR SECONDARY SCHOOLS

FORM FOUR

CHEMISTRY

QUESTIONS AND ANSWERS

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INTRODUCTION

This paper consists of chemistry series of examinations **(for form four students)**. All exams have been composed based on the current recommend format from National Examination Council of Tanzania (NECTA) for the year of 2022. Some exams have been collected and reorganized from regional mock exams conducted in the year of 2020 and 2021.

The paper Aim to exposure students to variety of questions and how to attempt questions properly. Also, will help students to adapt new format as recommended by NECTA toward their final examination.

Not only students but also teachers can use it as teaching and learning material during their teachings.

ACKNOWLEDGEMENT

First and foremost, we would like to **praise and thank God**, **the Almighty**, who has granted countless blessing, knowledge, and opportunity to the **Tzshule Team**, so that we have been finally able to accomplish the work.

Secondly, we would like to thank CEO & Founder of Tzshule (**Mr. George Ramadhani**) for not giving up on his dream of developing and providing educational materials electronically throughout the country.

We'd also like to thank all **Tzshule Users** for your support and positive feedback, since it motivates and encourage us to do more and more in developing educational materials.

- GOD MAY BLESS YOU ALL -

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QUESTIONS

THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM FOUR EXAMINATION - SERIES 01

CHEMISTRY

Time 3:00 Hrs **YEAR: 2022**

Instructions

1. This paper consists of section A, B & c with a total of fourteen (14) questions.
2. Answer all questions in section A & B and one (1) question from section C.
3. Cellular phones and any unauthorized materials are not allowed in the examination room.
4. Write your examination number on every page of your answer booklet(s).
5. The following constants may be used
   Atomic masses: H=1, O=16, S=32, Ca=40, Zn=65, Cl=35.5, Na=23, C=12, Ag=108

   Avogadro's number = 6.02×10^{23}
   GMV at s.t.p = $22.4dm^3$
   1 faraday = 96,500 coulombs
   1 litre = $1dm^3 = 1000cm^3$

SECTION A (15Marks)

Answer all questions in this section

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet provided.

   i. What number of faradays of electricity is required to deposit 4g of calcium from molten calcium chloride?

   A. 0.1 B. 0.2 C. 0.4 D. 0.3 E. 0.7

   ii. When methane undergoes substitution reaction with excess chlorine. What is the final product?

   A. Chloromethane
   B. Dichloromethane
   C. Trichloromethane
   D. Tetracloromethane
   E. Monochloromethane

iii. Which name is given to the uniform mixture of metals?

A. Compounds B. Alloys C. Ores D. Salts E. Mixtures

iv. The reaction between iodine and hydrogen is represented by the following equations

$I_{2(g)} + H_{2(g)} \rightleftharpoons 2HI_{(g)} \quad \Delta H = {}^{-}XKJ/mol$

The reaction is:-

A. An endothermic reaction

B. A displacement reaction

C. A neutralization reaction

D. A thermal decomposition reaction

E. An exothermic reaction

v. The electronic configuration of this element Cl^- is

A. 2:8:7 B. 2:8:8 C. 2:8:8:1 D. 2:8:8:2 E. 2:8:6

vi. Glucose dissolves in water to form a uniform mixture.

A. water is solution, glucose is solvent and the product is solute

B. water is solute, glucose is solvent and the product is solution

C. water is solvent, glucose is solute and the product is solution

D. water and glucose forms immiscible mixture

E. water and glucose form emulsion

vii. The reason why white anhydrous copper (II) Sulphate turns blue when exposed in Atmosphere is that it

A. Absorbs water vapour

B. Reacts with oxygen

C. Reacts with carbon dioxide

D. Become dry

E. Release water to the Atmosphere

viii. In order to produce the greatest amount of hydrogen in a short time, one gram of magnesiumribbon should react with

A. $10cm^3$ of 0.5M Sulphuric acid

B. $40cm^3$ of 0.5M acetic acid solution

C. $40cm^3$ of 0.5 Sulphuric acid solution

D. $20cm^3$ of 1M Sulphuric acid solution

E. $20cm^3$ of 1M acetic acid solution

ix. Which of the following substances represent a group of acid oxides?

A. Carbon dioxide, carbon monoxide and Sulphur dioxide.

B. Sulphur trioxide, Nitrogen dioxide and Nitrogen monoxide

C. Carbon dioxide, Sulphur dioxide and dinitrogen oxide

D. Sulphur trioxide, carbon dioxide and Nitrogen dioxide

E. Carbon monoxide, nitrogen oxide and sulphur dioxide.

x. Which action should be taken immediately after concentrated sulphuric acid is spilled onthe skin?

A. It should be rinsed off with large quantities of running water.

B. It should be neutralized with concentrated NaOH

C. The affected area should be wrapped tightly and shown to a medical health provider

D. It should be Neutralize with solid $CaCO_3$

E. It should be neutralized with concentrated KOH

2 Match the items in List A with the response in List B by writing the letter of the corresponding responses beside the item number.

LIST A	**LIST B**
i, Biogas	A Industrial gas which is a mixture of hydrogen and carbon monoxide
ii, Biomass	B Non- renewable gaseous fuel
iii, Natural gas	C Industrial gases fuel which is a mixture of nitrogen gas, hydrogen and carbon monoxide
iv, Producer gas	D Gases fuel derived from decomposition of biological waste
v, Water gas	E Renewable source of energy
	F Water gas and ammonia
	G Ammonia and butane

SECTION B (70Marks)

Answer all questions in this section

3. (a)

i. What is laboratory?

ii. Name two main compound used in preparation oxygen gas in laboratory

iii. Why hydrogen gas is collected by downward displacement method?

(b) Complete and balance the following chemical reaction

i. $NaOH_{(ag)} + HCl_{(ag)} \longrightarrow$

ii. $NH_4Cl_{(ag)} + NaOH_{(ag)} \longrightarrow$

iii. $Ca(OH)_{2(ag)} + H_2SO_{4(ag)} \longrightarrow$

iv. $Mg_{(s)} + HNO_{3(ag)} \longrightarrow$

4. Element R having atomic number 20 combines with element S having atomic number 17 to form a certain compound

a) Write the formula of the compound and state the type of bond formed in the compound

b) Give any three properties of the compound formed in 7(a) above

5. a) How many chlorine molecules are in $20cm^3$ of chlorine gas at s.t.p.

b) Calculate number of ions present in 5g of copper (II) Nitrate

6. a) State Le Chatellier's principle.

b) In the industrial preparation of Sulphur trioxide equilibrium is established between Sulphur dioxide and oxygen gas as follows:

$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)} \quad \Delta H = -94.9KJ/mol.$$

i. How would you adjust temperature and pressure to maximize the proportion of the productat equilibrium?

ii. Why is it unfavorable to work with very high pressure and very low temperature in thecontact process?

iii. What catalyst is used to speed up the rate of formation of Sulphur trioxide before attaining the equilibrium?

7. a) What is molecular formula?

b) You are provided with a compound composed of 22.2% Zinc, 11.6% Sulphur, 22.3% oxygen and the rest percentage is water of crystallization. Calculate the molecular formula of the compound if its molecular mass is 283.

8. In one titration experiment $25cm^3$ of hydrated sodium carbonate $Na_2CO_3.XH_2O$ were titrated against $20cm^3$ of 0.25M hydrochloric acid. The solution of sodium carbonate was made by dissolving 7.15g of the compound in $250cm^3$ of distilled water.

Use the above data to answer the following questions

a) Write down a balanced equation to represent the acid base reaction

b) i. Find the morality of the anhydrated sodium carbonate

ii. Calculate the value of X in the hydrated sodium carbonate.

9. a)Describe the use of each of the following during extraction of Sodium
   i. Calcium Chloride
   ii. Graphite anode
   iii. Steel gauze

   b).Why sodium is collected by upwards in the downs cell

   c). Write electrodes reaction in downs cell during extraction of Sodium

10. a) State Faraday's Laws of Electrolysis

    b) Dilute silver nitrate solution was decomposed by the passage of electric current through it. What mass of silver and what volume of oxygen (Measured at s.t.p) would be liberated in electrolysis by 9650 coulombs of Electricity?

11. a) In which other areas do we find the warning signs out of laboratory (give four point)

    b) Explain how measurements of volume differ when using measuring cylinder and burette

    c) It is recommend that laboratory apparatus should be properly washed or wiped after use, explain the significance for this when

    i. Measuring volume of liquids

    ii. Measuring mass of substance

12. a) State three main physical properties of water and show the usefulness of each property

    b) State four industrial application of electrolysis

SECTION C (15 MARKS)

Answer one (1) question in this section

13. a) How can you tell water is polluted? Give two ways

    b) We have coal at Kiwira in Mbeya region. Authorities in the government have allowed use of coal for domestic and industrial purpose. What warming can you raise concerning likely effects? Give five points

14. a) Define the following terms
    i. functional group
    ii. homologous series
    iii. isomerism

b) Write down the molecular structure and the IUPAC names of the isomer whose molecular structure is C_4H_{10}

c) By naming the reagents, stating the conditions whenever possible using a balances equation describe how ethene could be converted into

i. Ethane

ii. Chloroethane

iii. 1,2-dibromoethane

iv. Ethanol

QUESTIONS

THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM FOUR EXAMINATION - SERIES 02

CHEMISTRY

Time 3:00 Hrs **YEAR: 2022**

1. This paper consist of three sections, A, B and C with a total of **fourteen** (**14**) questions
2. Answer **all** questions in section **A** and **B** and **one** (1) question from section C
3. Read each question carefully before you start answering it.
4. Non-programmable Calculators may be used.
5. Write your **Examination number** on every page of your answer booklet(s)
6. The following constant may be used:

Atomic masses:

H=1; C=12; O=16; Cu=64; Ba=137; Na=23; Fe=56; N=14; K=39 S=32; Cl=35.5 Ag =108.

- Avogadro's number = 6.02×10^{23}.
- G.M.V at s.t.p = $22.4 dm^3$
- $1 litre = 1 dm^3 = 1000 cm^3$
- 1Faraday=96500coulombs
- Standard pressure=760mmHg
- Standard temperature=273K
- $1 litre = 1 dm^3 = 1000 cm^3$

SECTION A (15 MARKS)

Answer **all** questions in this section.

1. For each of the item (i) – (x) choose the correct answer from the given alternatives and write its letter beside the item number in the answer booklets provided.

(i) What will happen when yellow coloured flowers are placed in a gas jar containing Chlorine gas?

A. Flowers will dry
B. Flowers will change colour to pink
C. There will be decolourization of flowers
D. Flowers will shrink
E. The flower will turn blue.

(ii) A pupil put pieces of glass in water but later wanted to remain only with solvent, what processes can be suitable to obtain solvent

A. Filtration and condensation
B. Evaporation followed by decantation
C. Evaporation and condensation
D. Filtration
E. Condensation followed by centrifugation.

(iii) One of the following is a characteristic property of Non-metals.

A. They are electronegative D. They form anions by loss of electrons
B. They behave as reducing agent E. They are lustrous.
C. They form cation by gaining electrons

(iv) The three components of the fire triangle includes the following

A. Fuel ,Air and heat D. Fuel ,nitrogen and heat
B. Fuel, hydrogen and heat E. Fuel, heat and Carbon.
C. Fuel ,oxygen and heat

(v) The dispersed phase of one or more liquid(s) in another liquid is called

A. Emulsion C. An alloy E. Solution
B. Suspension D. Precipitation

(vi) Mariam mixed two solutions together in a boiling test tube at 20°C. After mixing the solution the new product was found to have a temperature of 18°C. This shows that the reactions is:-

A. An exothermic
B. neutralization
C. An endothermic
D. combustion
E. Displacement

(vii) The extraction of iron in the blast furnace gives slag as one of the products, this slag consists of:- A. Molten calcium oxide

B. Molten iron
C. Molten calcium siliconate
D. Molten iron oxide
E. Molten calcium silicate

(viii) In the formation of soil, some soils are younger than others. The main reason for this fact is

A. The climate of the area
B. The nature of parent rock material
C. Living organism
D. Time in which parent rock is exposed to weathering process
E. The wind blowing in the area.

(ix) Which of the following relates closely in a kinetic nature of matter,

A. Solids appear to be stationary but their particles keep on vibrating
B. At low temperature matter exist in gases form
C. Particles in gaseous state are very close to each other
D. Molecules in liquid state can move in all one fixed direction E. Particles in solid state are very far compared to gaseous state.

(x) Considering the Atomic properties of elements in the periodic table

A. The first ionization energy increase down a group
B. The Atomic size increases down the group
C. The metallic character increases down a group
D. The atomic radius increases down the group
E. The number of number of shells remains constant down the group.

2. Match the statement in list A with the respective stage of scientific method in list B by writing the letter of correct response beside the item number in the answer booklet provided.

LIST A	LIST B
(i) A statement of how the results relates to hypothesis (ii) A series of investigations (iii)A statement that identified an event, fact or situation (iv)A tentative explanation (v) A step in which the researcher explains the results.	A. Conclusion B. Data analysis C. Data collection D. Experimentation E. Hypothesis F. Observation G. Problem identification

SECTION B (70 Marks)

Answer **all** questions in this section

3. (a) Indicators behave different in acidic and basic solution. Using tabular form show how different indicators behaves in different solution

(b) With the aid of chemical equation explain the facts that dilute acids reacts with both metal and metal hydroxides

4. (a) A form two student wanted to explain the properties of the sub atomic particles of an atom. Show how he will have to clarify the properties in the class of 40 students

(b) Explain the Daltons contribution to Atomic structure.

5. (a) Metals are extracted from their ore. Using three important metals in the table , mention the substance(s) added apart from the important ore and suggest the function(s) of the substance added as in the table

Name of metal to extracted	Substance(s) added in the reaction	Function of substance
Sodium		
Aluminium		
Iron		

(b) Explain why metals higher in the series are extracted by electrolysis and not by chemical reduction

6. (a) An employer wanted to produce more Copper in electrolysis for exportation, but he didn't know what factors to consider so that he can produce more copper. As young scientist and 21th century experts with enough knowledge of electrolysis, explain three factors on which the amount of copper deposited will depend.

(b) If the amount of Copper that was required to be exported by the employer was 8kg, calculate the number of Amperes that would be used for 3hrs

7. (a) Nitrates and Carbonates behaves different on heating, using balanced chemical equation show how the nitrate and carbonate of the following element behaves on heating

(i) Sodium
(ii) Calcium
(b) Explain any three uses of the metal Nitrate

8. (a)Explain the concept of Safety measure for a Chemistry Laboratory

(b) Explain the significance of each of the following
(i) A chemistry Laboratory should have safety measure
(ii) It is necessary to familiarize yourself with Laboratory apparatus (iii) Chemicals in the laboratory must be labelled and stoppered.
(iv) All people working in the laboratory should wear appropriate protective clothes.
(v) Refrigerators and freezers used in the laboratory must be labeled *For chemical use only*

9. (a) A gift of 5g of Calcium Carbonate was given to John and 8g of Calcium Carbonate to Juma. The samples were stored in the same container when John and Juma arrived at home. Next day in the morning their sister Asha accidentally poured dilute hydrochloric Acid in the container and a colourles gas **R** was produced
(i) Suggest the name of the colorless gas **R**.produced
(ii) What chemical test will you conduct to identify gas **R**

(b) What volume of the gas **R** was produced
(ii) If the gas was collected by Asha, how would she use the gas .

10. (a) Ammonia is prepared in the Haber process according to the following reaction equation

$$N_2 + 3\,H_2 \rightleftharpoons 2NH_3 \quad \triangle \quad -192kjmol^{-}$$

If the reaction attains equilibrium explain

(i) Why increase in temperature of the equilibrium mixture decrease the yield of Ammonia
(ii) What will happen to the rate of production of Ammonia if Catalyst is added
(iii) Why increase in pressure of the equilibrium mixture would increase the production of Ammonia.

(b) Sketch an energy level diagram for the forward reaction in 10(a) above.

11. (a) Suppose you have been asked to mention the general components of soil at your school. What five components of the soil will you consider?
(b) What is the importance of soil texture to the growth of plants?

12. (a) State the importance of qualitative analysis in real life
(b) State important preliminary stages that you may involve when detecting salt sample.

SECTION C (15 Marks)

Answer **one** (1) question from this section

13. Suppose you have been invited in **Texas**-*The South Central region* of the United States to explain on how Sulphur can be extracted from the ground. With the aid of diagram, explain how you can apply the knowledge you have to obtain Sulphur from the ground.

14. Carbon exists in different allotropic form. Mention the allotropes and from each state the general properties and uses of each allotrope

QUESTIONS

THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM FOUR EXAMINATION - SERIES 03

CHEMISTRY

Time 3:00 Hrs **YEAR: 2022**

Instructions

1. This paper consists of sections A, B, and C with a total of **fourteen (14)** questions
2. Answer **all** questions in sections A and B and **one (1)** question from section C
3. Sections A and C carry **fifteen (15)** marks each and section B carries **seventy (70)** marks.
4. Cellular phones and any unauthorized materials are not allowed in the examination room
5. Write your **Examination Number** on every page of your answer booklet(s)
6. The following constants may be used.
    - Atomic masses: **H = 1, C = 12, N = 14, O = 16**, **Na = 23, S = 32, Ca = 40, Cl = 35.5, Cu = 64, Zn** = 65
    - Avogadro's number = **6.02×10^{23}**
    - GMV at s.t.p = **22.4 dm^3**
    - 1 Faraday = **96,500** coulombs
    - Standard pressure = **760** mmHg
    - Standard temperature = **273** K
    - **1** Litre = **1** dm^3 = **1000** cm^3

SECTION A (15 Marks)

Answer **all** questions in this section

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet provided

(i) The solution with pH of 5 is

A. A strong base
B. A neutral
C. A weak acid
D. A strong acid
E. A weak base

(ii) Alcohols react with carboxylic acids to form a group of organic compounds called

A. Alkynes
B. Haloalkanes
C. Esters
D. Alkenes
E. Alkanes

(iii) Which of the following is not a component of the first aid kit?

A. Goggles
B. A pair of scissors
C. Dropper
D. Gloves
E. Razor blade

(iv) A rapid chemical reaction that releases energy in form of light and heat is called

A. Combustion
B. Decomposition
C. Displacement
D. Neutralization
E. Precipitation

(v) The molarity of a solution containing 26.5g of anhydrous sodium carbonate in 5 dm^3 of the solution is……..M

A. 0.05
B. 0.25
C. 1.25
D. 5.3
E. 0.025

(vi) Why oxygen differs from other gases?

A. It neither burns nor support combustion
B. It supports combustion but does not burn
C. It burns but does not support combustion
D. It burns and supports combustion
E. It explodes and support combustion

(vii) The oxidation state of chlorine in sodium chlorate ($NaClO_3$) is:

A. -1
B. +2
C. +5
D. +3
E. -3

(viii) Elements lose or gain electrons to form:

A. Isotopes
B. Radicals
C. Molecules
D. Ions
E. Allotropes

(ix) Insoluble salts like barium sulphate, generally can be obtained in the laboratory by:

A. Evaporation of its concentrated solution
B. Crystallization
C. Precipitation
D. Decomposition
E. Displacement reaction

(x) In a blast furnace carbon monoxide is prepared by passing carbon dioxide over a red hot coke. Carbon dioxide is:

A. An accelerator
B. An oxidizing agent
C. A reducing agent
D. A catalyst
E. Oxidized

2. Match the items in **list A** with the responses in list B by writing the letter of the correct response beside the item number in the answer booklet provided.

List A	**List B**
(i) Isomerism	A. Soil pollution, nutrient pollution
(ii) Bonding in molecules of nitrogen	B. Compounds with the same molecular formula but different structural formulae
(iii) Ammonia in water	C. Presentation of reactants and product
(iv) Chemical equation	D. Pungent choking smell
(v) Eutrophication	E. Triple bonds
	F. Bleaching
	G. Ionization

SECTION B (70 Marks)

Answer **all** questions in this section

3. (a) Giving an example for each, give four uses of matter in daily life
(b) Why are chemical symbols useful in Chemistry? Give three reasons. **(7 marks)**

4. (a) Distinguish between temporary hardness and permanent hardness of water

(b) By use of equations, show how each of the type of hardness in (a) above can be eliminated. **(7 marks)**

5. (a) Define the following terms

(i) Acid

(ii) Base

(iii) Salt

(b) State any four importance of neutralization in daily life. (**7 marks)**

6. (a) Briefly explain three importance of chemical equation

(b) Give a balanced chemical equation between the reaction of sodium carbonate and hydrochloric acid. **(7 marks)**

7. (a) Calculate the concentration in g/dm^3 of vinegar (CH_3COOH) if 25.0 cm^3 of 0.1 M sodium hydroxide reacts with 12.5 cm^3 of vinegar.

(b) By giving a reason, suggest the suitable indicator for the reaction in (a) above. **(7 marks)**

8. (a) Explain the meaning of molar volume of a gas

(b) If 0.5g of hydrogen gas is exploded in air, what is the mass of water formed? **(7 marks)**

9. (a) Ammonia gas is manufactured by nitrogen gas with hydrogen gas in the presence of a catalyst. Write a balanced chemical equation for the reaction and explain the role played by the catalyst in this reaction.

(b) When hydrogen sulphide gas is passed through sulphur dioxide a yellow deposit of sulphur is produced immediately. Write the chemical equation for this observation. **(7 marks)**

10. (a) Give the meaning of the following terms

(i) Cracking

(ii) Isomerism

(b) Write the structures of the following

(i) 2,2-dichloro-3-methylpentane

(ii) 4-ethyl-2,6-dimethylheptane. **(7 marks)**

11. (a) Element X has 20 electrons and mass number of 40. Work out the number of each type of nucleons present.

(b) Calculate the empirical formula and molecular of the compound having a relative molecular mass of 76 containing 15.8% of carbon and 84.2% of sulphur. **(7 marks)**

12. (a) Arrange the following elements in the order of increase in electronegativity: F, Br, I, N, C and Cl

(b) Predict the products in the following reactions and give reason if any:

(i) $Cl_2(g) + 2KI\ (aq) \longrightarrow$

(ii) $Br_2\ (g) + NaCl\ (aq) \longrightarrow$

(iii) $Cl_2\ (g) + HF\ (g) \longrightarrow$ **(7 marks)**

SECTION C (15 Marks)

Answer **one (1)** question in this section

13. Explain how soil fertility can be lost by erosion, water logging, leaching (flooding) and burning. **(15 marks)**

14. What are protective and remedial measures of water pollution? **(15 marks)**

QUESTIONS

THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM FOUR EXAMINATION - SERIES 04

CHEMISTRY

Time 3:00 Hrs **YEAR: 2022**

INSTRUCTIONS

1. This paper consists of sections A, B and C with a total of **fourteen (14)** questions.
2. Answer all questions in section A and B and only one question in section C
3. Calculator, cellular phones and any unauthorized materials are not allowed in the examination room
4. Write your examination number on every page of your answer book sheets
5. The following constant may be used:

Atomic masses: H=1, Cl=35.5, K=39, O=16, Na=23, Ba=137, N=14 Al=27, S=32, Pb=207, Na= 23, C=12, H=1, K=39, Mn= 55, Mg=24

- Avogadro's number = 6.02×10^{23} Particles
- 1 Faraday = 96500C
- Standard Pressure = 760mmHg
- Standard Temperature = 273K
- 1litre= $1dm^3$=$1000cm^3$=1000ml
- GMV at s.t.p =$22.4dm^3$

SECTION A (15 Marks)

Answer **all** questions in this section

1. For each of the items (i)-(x), choose the correct answer among given alternatives and write its letter beside the item number in the answer sheet provided

(i) Most laboratory apparatus are made up of glass because:

A. Is transparent
B. Is unreactive
C. Is readily available
D. Has high melting point
E. Stores chemical for quite long

(ii) The empirical formula differs from the molecular formula in a way that;

A. Molecular formula may have higher mass
B. Empirical formula may have higher mass
C. Always the empirical formula has lower mass
D. Always the molecular formula has higher mass
E. Empirical formula shows the simple ratio by mass while molecular formula shows actual number of atoms.

(iii)If chlorine water is exposed to bright sunlight for some time bubbles of gas are produced, the gas is;

A. Hydrogen
B. Hydrogen chloride
C. Oxygen
D. Chlorine
E. Water gas

(iv) During chemical reaction when the energy required to break bonds is lower than the energy released during bond formation; that reaction will be termed as:

A. Synthesis
B. Endothermic
C. Double decomposition
D. Exothermic
E. Explosion

(v) Water contain calcium hydrogen carbonate salt can be soften by :

A. Decantation
B. Reacting it with acid
C. Filtration
D. Neutralization
E. Using washing so

(vi) Which of the following organic compound do not belong to the homologous series alkanes?

A. CH_4
B. C_2H_2
C. C_4H_{10}
D. C_3H_8
E. C_8H_{18}

(vii) When a metallic element burns in air the compound formed is likely to be:

A. Basic oxide
B. Common oxide
C. A salt
D. Acidic oxide
E. A gas

(viii) In which state of matter does a substance have fixed volume and indefinite shape?

A. Liquid
B. Solid
C. Gaseous
D. Brownian
E. Excited

(ix) In sodium chloride, Sodium and chloride ions are held together by:

A. Covalent bond
B. Electrovalent bond
C. Dative bonds
D. Hydrostatic forces
E. Weak force of attraction

(x) The volume occupied by 8.6g of carbon dioxide at S.T.P is

A. 44.8 dm^3
B. 22.4dm^3
C. 48.4dm^3
D. 10.2dm^3
E. 4.4dm^3

2. Match the item In LIST A with the responses an LIST B by writing the letter of the correct response beside the item number in the answer booklet provided

LIST A	LIST B
i. Alkaline gas	A. Zinc
ii. Extracted by frasch process	B. Iron
iii. Quick lime	C. Ammonia
iv. Has variable oxidation state	D. NaOH
v. Its oxide is yellow when hot and white when cold	E. Sulphur
	F. Sulphuric acid
	G. Copper
	H. Calcium oxide
	I. Calcium hydroxide
	J. Sulphur dioxide

SECTION B: (70 marks)

Answer all questions in this section

3. (a) With one (i) example in each case, briefly explain the following

(i) Esterification reaction
(ii) Substitution reaction
(iii)Decomposition reaction

(b) Compound M has molecular mass of 84g. It has the following percentage composition by mass; 28.6% Magnesium, 14.3% Carbon and the rest is oxygen.

(i) Find the molecular formula of compound M
(ii) Give the chemical name for compound M

4. (a) What is the different between molecular equation and ionic equation? Give example

(b)Write balanced chemical equation for the reaction between copper and hot concentrated sulphuric acid

(c) Write balance ionic equation for the following reaction.

(i) A reaction between barium sulphate solution and sodium chloride solution to form barium chloride and sodium sulphate.
(ii) A reaction between zinc granules and dilute sulphuric acid to form zinc sulphate and hydrogen

5. (a) Define the following terms
(i) Electrochemical series
(ii) Anode
(b) An electric current of 2 amperes was passed through a solution of silver nitrate solution for an hour and one minute.

(i) Write the ionic equations for the reaction at both electrodes
(ii) Calculate the mass of the metal deposited at cathode.

6. (a) Explain briefly why alminium metal is used in
(i) Making aluminium windows
(ii) Making electrical cables
(iii)Making cooking utensils
(b) (i) Define the term dilution

(ii) What volume of 18.4M sulphuric acids is required to prepare $200cm^3$ of 0.5M of Sulphuric acid?

7. (a) Define the following terms

(i) Molarity

(ii) Concentration

(b) A mass of 14.3g of hydrated sodium carbonate solution ($Na_2CO_3.XH_2O$) was dissolved in water and diluted to make $250cm^3$ of solution. Then $25cm^3$ of the solution was titrated with 0.5M Hydrochloric acid. The volume of acid used was $20cm^3$. Determine the number of molecule of water of crystallization(X) in ($Na_2CO_3.XH_2O$)

8. Element X, Y and Z have atomic number 17, 18 and 20 respectively.
(a) (i) Write the electronic configuration of each element.

(ii) Basing on reactivity; which of the above element is likely to be found in uncombined state?

(b) (i) Write down three (3) characteristic of a bond formed between X and Z

(ii) Write the chemical formula of a compound formed when X combines with Z.

(iv)Name the type of bond formed when X and Z combine.

9. (a) (i) What is a homogenous mixture?

(ii) Is air a compound Or mixture? Give four reasons to support your answer

(b) Briefly state the stages of extraction of iron

(i) (c) Explain the role played by each of the following in the extraction of iron.

(i) Limestone

(ii) Hot air

(iii)Coke

10. (a) Explain the meaning of each of the following terms;

(i) Reversible reaction

(ii) Dynamic equilibrium

(iii)Chemical kinetics

(b) The industrial oxidation of sulphur dioxide is summarized in the equation below.

$2SO_{2\,(g)} + O_{2(g)} \leftrightharpoons 2SO_{3\,(g)}$ H = -94.9Kjmol^{-1}

What will be the effect of each of the following on the production of sulphur trioxide?

(i) Increase in moles of sulphur dioxide

(ii) Increase in pressure

(iii) Decrease of temperature

(iv) Decrease of moles of sulphur dioxide

11. (a) Define the following terms

(i) Cracking

(ii) Additional reaction

(iii)Fractional distillation

(iv)Unsaturated hydrocarbon

(b) The following names have been written wrongly according to the IUPAC system. Write the

structural formula of each compound from their wrong names and write the correct name

of each compound according to the I.U.P.A.C

(i) 1,1,1,1,-Tetramethylmethane

(ii) 4 – methyl pentane

(iii)2 – ethylpent – 4 – yne

(c) Complete the following organic reactions

(i) $CH_4 + Cl_2 \xrightarrow{uv.light}$

(ii) $C_2H_5OH \xrightarrow{conc.H_2SO_4}$

12. The following list of elements is arranged in the order of their reactivity with the most reactive taking the first position and the least reactive taking the least position

    Na, Ca, Mg, Zn, Fe, H, Cu, Hg.

    From this list of elements answer the following questions.

(a) Name the element that;
    (i) Is not found free in nature
    (ii) Is found free in nature
    (iii) Whose carbonate do not decompose when heated
    (iv) Can display both metallic and non- metallic characteristics

(b) Name the metal which reacts with
    (i) Cold water
    (ii) Steam but does not react with cold water

(c) Name an element whose oxide decompose on heating

(d) An element which cannot displace hydrogen from the dilute acid

(e) Write a balanced chemical equation for the reaction
    (i) Metal in **question12b(i)** react with cold water
    (ii) Oxide of element in **question12 (c)** is strongly heated

(f) Why hydrogen gas is included in the activity series of the metal?

SECTION C: (15 MARKS)

Answer only ONE question in this section

13. The environment supports life of every organism. Pollution of the environment has led to many catastrophic effects particularly to aquatic life. Describe aquatic pollution by analyzing its **five** causes, **five** effects and **four** remedial measures taken against it.

14. Write an essay on soil fertility by using the following guideline;
    (i) Meaning of soil fertility
    (ii) Methods used to maintain soil fertility (**four (4)** methods).
    (iii) How does soil lose its fertility (**four (4)** ways)
    (iv) A fertile soil is not always productive. Justify the statement by explaining **five** factors affecting soil productivity.

QUESTIONS

THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM FOUR EXAMINATION - SERIES 05

CHEMISTRY

Time 3:00 Hrs **YEAR: 2022**

INSTRUCTIONS

1. This paper consists of sections A, B and C with total of 14 questions.
2. Answer all questions in sections A and B and **one (1)** question from section C.
3. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
4. Write your **examination number** on every page of your answer sheet(s)
5. The following constants may be used
   - Atomic masses: H=1, O=16, N=14, S=32, Zn=65, Cl=35.5, Cu=64
   - Avogadro's constant $=6.02 \times 10^{23}$
   - GMV at s.t.p=22.4 dm^3
   - 1 Faraday =96500 coulombs
   - Standard pressure = 760 mm Hg
   - Standard temperature =273 K
   - 1 litre=1 dm^3

SECTION A: 15 MARKS
Answer all Questions

1) For each of the items (i) – (x) choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet provided.

i. A chemist is a person who investigates and collects data on changes of chemical substances before making any conclusion. Which important qualities must this person have?
   A. Mastery of the periodic table of elements
   B. Skills in working report in English language
   C. Keen in the use of most of the sense organs
   D. Interact in the use of computer
   E. Familiar with other science subject

ii. A mixture of oil and water is termed as:
   A. Suspension
   B. Residue
   C. Filtrate
   D. Solution
   E. Emulsion

iii. When a nonmetal ion become an atom it:
   A. Gains electron(s)
   B. Gains proton(s)
   C. Loss proton(S)
   D. Loose electron(s)
   E. Gains and loose electron(s)

iv. "Water is referred to as the universal solvent" what does this mean?
   A. Water is neither acidic nor basic
   B. Water exist as three states of matter
   C. Water dissolves both organic and inorganic solvent
   D. Water is used more in domestic
   E. Water dissolves more substance

v. The volume of 0.2M H_2SO_4 required to neutralize completely 25.00 cm^3 of 0.05M KOH is:
   A. 0.626 cm^3
   B. 6.125 cm^3
   C. 6.315 cm^3
   D. 3.125 cm^3
   E. 12.5 cm^3

vi. The property of metals that can be turned into thin sheets is called?
   A. Ductility
   B. Sonorosity
   C. Malleability
   D. Expansibility
   E. Lustrous

vii. One among the following substances is made by application of chemistry in agriculture:
   A. Yeast
   B. PVC
   C. Toothpaste
   D. Pesticides
   E. Alcohols

viii. Which of the following compounds does not belong to the alkane homologous series?
   A. C_2H_4
   B. CH_4
   C. C_4H_{10}
   D. C_3H_8
   E. C_5H_{12}

ix. The IUPAC name of H_2SO_4 is:
   A. Sulphuric (IV) acid
   B. Sulphuric acid
   C. Sulphuric (V) acid
   D. Sulphorous acid
   E. Sulphuric (VI) acid

x. Which oxides are gaseous at room temperature?
   A. Carbon dioxide and copper(II) oxide
   B. Sulphur dioxide and Copper (I) oxide
   C. Carbon dioxide and Sulphur dioxide
   D. Copper oxide and Iron oxide
   E. Iron oxide and Carbon dioxide

2. Match the item in list A with the responses in list B by writing the letter of the most correct response besides the item number in the answer sheet provided.

LIST A	LIST B
i. Synthesis reaction ii. Precipitation reaction iii. Redox reaction iv. Displacement reaction v. Neutralization reaction	A. A single compound breaks down into two or more simpler substance. B. Reaction between acid and base. C. Two or more reactants form one compound. D. A more reactive element displaces a less reactive element from a compound. E. Reaction that involves a transfer of electrons between two species. F. Formation of an insoluble salt in which two solutions containing soluble salt are combined. G. Reaction between alcohol and carboxylic acid.

SECTION B (70 Marks)
Answer all questions in this section

3. (a) Explain briefly the following terms
   i. Mole
   ii. Molar volume

   (b) A current of 0.2A is passed for exactly 30 minutes resulting in the deposition of 0.4g of silver.
   i. Briefly state the five factors which may affect the amount of silver deposited
   ii. Calculate the quantity of electricity passed through the electrolyte
   iii. Calculate the number of electrons passed through the electrolyte **(7marks)**

4. (a) Distinguish bases from acid based on their physical properties. Two points.

   (b) Briefly give one reason for the following uses of metals
   i) Aluminum is used for making window frame.
   ii) Iron is used for making bridges.
   iii) Copper for electrical wiring.
   iv) Lead for roofing.
   v) Zinc for coating of iron. **(7marks)**

5. (a) The formation of lime scale in water pipes has both an advantage and a disadvantage. Explain

(b) With the help of the chemical questions, explain how you can remove temporary and permanent hardness of water. **(7marks)**

6. The industrial preparation of ammonia in harber process is represented by the following equation:

$$N_{2(g)} + 3H_{2(g)} \xleftrightarrow{catalyst} 2NH_{3(g)} \quad \Delta = -46.2\text{KJ/mol}$$

Study the equation carefully and answer the following questions that follow. What will happen to the position of equilibrium if:

a) (i) The temperature of the system is increased?

(ii) More nitrogen gas is added to the equilibrium mixture?

(iii) The formed ammonia gas is removed from the equilibrium mixture?

b) (i) What is the use of catalyst in the reaction above?

(ii) What is the meaning of the negative sign against the value of heat change-46.2kj/ mol in chemical reaction given above?

(iii) List down three (03) factors which affect the position of equilibrium in the reversible reaction. **(7marks)**

7. (a) An atom M has an atomic number 15 and mass number 31.

(i) What is the number of protons and neutrons?

(ii) Write the electronic configuration of atom M

(b) What amount of current in ampere will be liberated within one hour if 2.24 dm^3 of chlorine were produced at S.T.P at the anode, when sodium chloride solution is electrolyzed? **(7marks)**

8. (a) Using balanced chemical equation, show steps on how methane undergoes complete chlorination in the presence of sunlight.

(b) Write the structures of the following organic compounds

(i) 1,2_dichloroethane

(ii) 2,2_dimethylpropane

(iii) pent-2-yne **(7marks)**

9. (a) Give two reasons which verify the statement that "chemists prefer to use molecular equations rather than word equations.

(b) Dilute hydrochloric acid is reacted with zinc granules the product formed is zinc chloride and hydrogen gas is evolved.

(i) Write the balanced chemical equation for the reaction.

(ii) Write the ionic equation for the reaction. **(7marks)**

10.(a) (i) Write a balanced chemical equation for the laboratory preparation of chlorine gas

(ii) Why is the gas prepared in the fume chamber?

(b) (i)Why the chlorine can be collected over water?

(ii) List down two uses of chlorine gas **(7marks)**

11.(a) Write the chemical symbols of the following elements

Silver, mercury, lead, beryllium and iron

(b) You are provided with compound composed of 22.2% zinc, 11.6% Sulphur, 22.3% oxygen, and the rest percentage is water crystallization. Calculate the molecular formula of the compound if its molecular mass is 283. **(7marks)**

12. (a) Write down five characteristics of a good fuels

(b) Why we are advised to use alternative source of energy such as solar energy, biogas and wind energy rather than using fossils fuels source of energy. Give two points. **(7marks)**

SECTION C (15 marks)

Answer one (1) question from this section

13.(a) Explain the effects of the following to the environment:

i) Acidic rain

ii) Artificial fertilizers

iii) Plastic containers

(b) i) What control measures should be taken to prevent the destruction of ozone layer.

ii) List down three gases that cause greenhouse effects.

14. Describe four common stages for the extraction of metals and give reasons why extraction of gold does not follow all four stages?

QUESTIONS

THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM FOUR EXAMINATION - SERIES 06

CHEMISTRY

Time 3:00 Hrs **YEAR: 2022**

INSTRUCTIONS

1. This paper consists of three sections A, B and C with a total of **fourteen (14)** questions.
2. Answer **all** questions in section A and B and choose one question from section C.
3. Cellular phones and any unauthorised material are not allowed in the examination room.
4. Write your examination number on every page of your answer sheet (s).
5. The following constants may be used:
   - ✓ Atomic masses: H = 1, O = 16, C =12, Na = 23, Al = 26,K = 39,Cu = 63.5, Cl = 35.5
   - ✓ Avogadro's constant = $6.02 \times 10^{23.}$
   - ✓ GMV =$22.4dm^3$
   - ✓ 1litre= $1dm^3$ =$1000\ cm^3$

SECTION A (15 Marks)
Answer all questions in this section

1. For each of the items (i) – (x) choose the correct answer among the given alternatives and write its letter beside the item number in your answer sheet provided.

   (i) If the results you obtain from an experiment do not support your hypothesis
   - A. Change your experiment
   - B. Leave out the results
   - C. Give ideas for further testing to find a solution
   - D. Identify a new problem
   - E. Change dependent variable

   (ii) Which of the following sets of processes uses a gas that ignites with a pop sound when a lighted sprint is passed through it?
   - A. Balloon filling, welding and diving
   - B. Hardening oil, balloon filling and welding
   - C. Hardening oil, balloon filling and diving
   - D. Fueling rocket, diving and welding
   - E. Balloon filling, fueling rocket and diving

   (iii)One advantage of hard water is that it
   - A. Causes the accumulation of salts in water pipes leading to blockage
   - B. Causes increased tooth decay
   - C. Uses more soap to lather
   - D. Contains minerals which are useful to the body
   - E. Causes the formation of lime scale in the boilers

(iv) A and B are elements in the same period of the periodic Table. A is in group II and B is in group III. Which of the following statements is not true about the two elements?

A. B has one proton more than A in the nucleus
B. The atomic number of B is one unit greater than that of A
C. A has one electron less than B in its outermost shell
D. A has one electron more than B in its outermost shell
E. A and B have the same number of shells

(v) An example of a salt which is insoluble in cold water but can dissolve in hot water is

A. Sodium chloride
B. Lead chloride
C. Calcium carbonate
D. Silver chloride
E. Copper carbonate

(vi) Which state is involved when drying wet clothes?

A. Liquid to solid
B. Solid to gas
C. Gas to liquid
D. Liquid to gas
E. Gas to solid

(vii) The formula below represents some chemical substances. Which formula represents a substance that does not contribute to global warming?

A. H_2
B. CH_4
C. CO_2
D. N_2O
E. SO_2

(viii) An alkyl group has a general formula

A. C_nH_{2n-1}
B. C_nH_{2n}
C. C_nH_{2n+2}
D. C_nH_{2n+1}
E. C_nH_{2n-2}

(ix) The empirical formula of a certain compound is CH_3. Its molar mass is 30g what will be its molecular formula?

A. CH_4
B. C_2H_4
C. C_2H_6
D. C_2H_8
E. C_4H_{12}

(x) What type of fire is associated with electrical equipment?

A. Class E
B. Class F
C. Class B
D. Class C
E. Class A

2. Match the items in List A with the responses in List B by writing the letter of the correct response beside the item number in the answer sheet.

List A	List B
(i) A solvent which dissolves most substances to form solutions. (ii) A substance that has no definite shape or size. (iii) A substance that has a fixed shape and volume (iv) A substance whose components can be separated by physical means. (v) Homogeneous mixture of two or more substance	A. Solid B. Solution C. Water D. Sugar E. Milk F. Gas G. Liquid H. Air

SECTION B (70 Marks)
Answer all questions in this section

3. (a) Why should a good fuel have the following characteristics?
   (i) High energy value
   (ii) Affordable
   (iii) Low content of non – combustible material
   (b) Briefly explain how Chemistry is put into use in your community (Give two points)

4. (a) State one use of each of the items (i) – (v) in administering First Aid.

S/N	Item	Use
(i)	Soap	
(ii)	Bandage	
(iii)	Sterile gauze	
(iv)	Iodine tincture	
(v)	Petroleum jelly	
(vi)	Cotton wool	

   (b) Aluminium reacts with oxygen to form aluminium oxide. How many grams of potassium chlorate will be heated to produce enough oxygen to form 5.1g of aluminium oxide?(***7 marks***)

5. (a) By using the idea of particles , explain why:
   (i) It is easy to pour a liquid
   (ii) A gas will completely fill any container
   (iii) A solid expands when it is heated
   (b) With the aid of a chemical equation, briefly explain how:
   (i) Temporary hardness of water can be removed by boiling
   (ii) Permanent hardness of water can be removed by chemical means
   (7 marks)

6. (a) (i)Show and name a type of a chemical bond found between fluorine atoms in a fluorine molecule.
   ii)Name other type(s) of Chemical bond formed by fluorine with other elements. Give an example of a compound in which fluorine forms this type of bond
   (b) Differentiate between dilute Nitric acid and dilute phosphoric acid(**7marks)**

7. (a) i) Suggest two ways in which a luminous is unsuitable for heating in the laboratory
ii) Under what conditions a non – luminous flame is obtained from a Bunsen Burner?
(b) Calculate the time in minutes required to deposit 1.184 grams of copper if a current of 2 amperes was used. **(7marks)**

8. (a) i) State two physical properties of water
ii) A certain student at Mwembeni Secondary School found three gas jars named A,B and C inside their School laboratory, one of the gas jar contains Oxygen gas, How could the student identify the Oxygen gas?

(b) A 250cm^3solution of diluted Hydrochloric acid was prepared from 10 cm^3 of a stock solution of dilute hydrochloric acid of unknown concentration. 25 cm^3 of the prepared acid was completely neutralised by 12.5 cm3 of 0.5M aqueous Sodium hydroxide. What is the concentration of the stock solution? **(7marks)**

9. (a) Draw electronic diagram to represent:
i) Chlorine atom
ii) Chlorine ion
(b) i) Identify two major crystalline allotropes of carbon
ii) Which one between the two allotropes in (b) (i) conducts electricity? Give a reason. **(7marks)**

10. A certain metal Z is found in common salt and stored under kerosene due to its reactivity, but also extracted by electrolytic reduction from its ore.
(a) (i) identify the name of the cell for the extraction of metal Z.
(ii) Why metal Z cannot be extracted from aqueous solution of its ore?
(b) Describe the type of reaction of the following chemical equations
(i) $3KOH_{(aq)} + FeCl_{3\,(aq)} \longrightarrow 3KCl_{(aq)} + Fe(OH)_{3(s)}$
(ii) $Ca(NO3)_2\,(s) \longrightarrow CaO_{\,(s)} + NO_{2\,(g)} + O_{2\,(g)}$ **(7marks)**

11. Consider the reaction below:
$A_2(g) + B_2\,(g) \longrightarrow 2AB\,(g) \qquad H = -55KJmol^{-}$
(i) With a reason state the type of the reaction in consideration of heat changes
(ii) If the activation energy is XKJ draw the energy level profile for the reaction **(7marks)**

12. (a) Identify the substance by using the following information:
i) A solid is yellow when hot and white when cold
ii) When water is added to a white powder heat is evolved and the white powder changes to blue crystals.
iii) An aqueous solution of a greenish crystalline sulphate forms a pale –green precipitate with sodium hydroxide solution which turns to brown on standing and when exposed to air.
(b) With the help of chemical equations explain what happens to the following compounds of ammonia when heated in separate test tubes:
i) A mixture of ammonia chloride and sodium hydroxide solution
ii) Ammonium chloride crystals **(7marks)**

SECTION C (15 Marks)
Answer only one question from this section.

13. Explain how to handle chemicals having the warning signs of flammable, corrosive, harmful, explosive and toxic in the laboratory.

14. Using five examples, explain how the process of neutralization is important in day to day life.

QUESTIONS

THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM FOUR EXAMINATION - SERIES 07

CHEMISTRY

Time 3:00 Hrs **YEAR: 2022**

INSTRUCTIONS

1) This paper consists of section A, B and C with total of fourteen questions
2) Answer all questions from section A and B and only One (1) question from section C
3) All writings should be in blue or black Ink pen excepts for diagrams that must be drawn in a pencil
4) Cellular phones and any unauthorized materials are not allowed in the examination room
5) Write your Examination number (index number) on every page of your answers booklets provided
6) The following constant may be used
   Atomic masses: H = 1, O=16, N=14, S=32, Ag=108, and CU = 63.5
   - Avogadro's number = $6.02 \times 10^{23} mol^{-1}$
   - G.M.V at s.t.p = 22.4 dm^3
   - 1 faraday = 96500 coulombs
   - Standard temperature = 273K
   - 1 litre = $1dm^3 = 1000cm^3$

SECTION A {15 Marks}

Answer all questions in this section

1. For each of the following items (i – x) choose the correct answer from among the given alternatives and write its letter besides the item number in the answer booklet(s) provided.

   i. A simple proof that some chemical reactions takes place in our body is that
   - **A.** Doctors tell us so in hospital
   - **B.** We eats a balanced diet
   - **C.** We occasionally fall sick
   - **D.** The waste products form our bodies differs from the food we eats
   - **E.** There is no proof

   ii. Why is hydrogen gas collected over water and by upward delivery method?
   - **A.** It is soluble in both air and water
   - **B.** It is soluble in water and less denser than air
   - **C.** It is insoluble in water and denser than air
   - **D.** It is insoluble in water and denser than air
   - **E.** It is soluble in water and denser than air

iii. An a aqueous solution turn red litmus paper into blue, excess addition of which solution would reverse the change:

A. Baking powder
B. Aqueous ammonia
C. Quick lime
D. Lime
E. Vinegar

iv. If 0.5g of hydrogen gas is exploded in air, the mass of water formed is;

A. 0.75g
B. 18g
C. 40g
D. 1.8g
E. 4.5g

v. Brown ring and copper turning tests

A. For nitrates
B. For sulphates
C. For carbonates
D. For chlorides
E. For hydrogen carbonates

vi. Three elements X, Y and Z are in the same period of the periodic table. The oxide of X is amphoteric, the oxide of Y is basic and the oxide of Z is acidic. Which of the following shows the arrangement of element in order of increasing atomic number:

A. X,Y,Z
B. Y,X,Z
C. Z,X,Y
D. Y,Z,X
E. Z,Y,X

vii. The Oxidation state of chlorine in potassium chlorate is

A. $^{+}1$
B. $^{+}7$
C. $^{-}1$
D. $^{+}5$
E. $^{+}6$

viii. If water does not easily form lather with soap, it is because of the presence of:

A. Sodium and calcium salts
B. Ammonium and aluminium salts
C. Calcium and magnesium salts
D. Zinc and aluminium salts
E. Iron salts

ix. What happens when an elements loose or gain an electron:

A. Isotopes are formed
B. Allotropes are formed
C. Molecules are formed
D. Ions are formed
E. Radicals are formed

x. The following reaction, $4NH_{3(g)} + 5O_{2(g)} \rightarrow 4NO_{(g)} + 6H_2O_{(g)}$ is an example of a:

A. Redox reaction
B. Neutralization reaction
C. Esterification reaction
D. Combination reaction
E. Decomposition reaction

2. Match the items in **List A** by choosing the correct answer form **List B** and write by writing the corresponding letter in box provided

LIST A	**LIST B**
i. Removing an electron form an atom or ion	**A.** Electronegativity
ii. A minimum energy for removing an electron form an atom or ion	**B.** Ionization energy
iii. A positive or negative charge	**C.** Atoms
iv. A tending of an atom to attract electrons	**D.** Atomicity
v. A tendency of an atom to lose electrons	**E.** Electropositivity
	F. Ionization
	G. Ions

SECTION B (70 Marks)

Answer all questions in this section

3. (a)The table below shows two brands of bottled water for drinking and the concentration of different mineral ions in each brand. Study the table and answer the questions below it.

Composition in mg/litre	**Mineral**	**Na^+**	**Ca^{2+}**	**Mg^{2+}**	**Cl^-**	**NO_3^-**	**SO_4^{2-}**	**Fe^{2+}**	**F^-**
	Uhai	40.0	3.05	4.15	14.18	0.48	10.0	0	1.76
	Dasani	22.32	2.69	0.11	6.50	1.0	8.0	0	0.45

i. Which brand of water is more hard: Explain
ii. State the benefit of Taking Calcium contained water
iii. Tap water is usual treated before being used. State what is added to perform that function

(b) Hydrogen and phosphorus are non metallic elements
i. Which one among the two is more electropositive
ii. Show your work clearly, write the chemical formula and name the compound formed when the two atoms combine

4. (a) Which conditions are required for rusting
(b) Suggest one best method for separating each of the following mixture
i. Common salt and water
ii. Iodine and sand
iii. Olive oil and water
iv. Ethanol and water

5. (a) With the aid of balanced chemical equation, briefly explain what happen when a crushed eggs shell is reacted with hydrochloric acid
(b) Name two main products formed when
i. Hot concentrated sulphuric acid is added to sugar cane
ii. Hot concentrated sulphuric acid is added to ethanol

6. (a) Explain why people suffering form heart burn usually uses woodashes for relief.
(b) $25cm^3$ of 0.1M hydrochloric acid were neutaliged by $23cm^3$ of sodium hydroxide solution. Calculate the concentration of an alkali in grams per litre

7. (a) Outline any three uses of carbondioxide gas in ammal and plants
(b) Complete the following reactions
i. $CH_2 = CH_2 + O_2 \rightarrow$
ii. $CaC_2 + H_2O \rightarrow$
iii. $CH_4 + Cl_2 \rightarrow$
iv. $C_2H_5OH + O_2 \rightarrow$

8. (a) How can the rate of chemical reactions be increased? Describe at least four ways
   (b) Study the following reaction

   $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)} \quad \Delta H = -46.2KJmol^{-}$

   Using Le chatelier's principle, suggest how you would use temperature and pressure to obtain the highest production of ammonia at equilibrium

9. a) i/ What is the aim of having fume chamber in the laboratory?
   ii/ Why the door in the laboratory opens outward?
   (b) State the uses of the following

Item	Use
i. Cotton wool	
ii. Petrol jelly	
iii. Pain killer	
iv. Bandages	
v. Razor blade	

10. (a) Giving reasons, explain why air is said to be a mixture of gases (Three reasons)
    (b) A current of 0.5A were made to flow through silver voltammeter for 40 minutes. Calculate the mass of silver deposited and equivalent weight of silver.
11. (a) Give the meaning of the following
    i. Soil PH
    ii. Liming
    (b) A certain compound contains 20% by mass of Mg. 26.5% by mass of sulphur and Z% of oxygen
    Find
    i. The value of Z **ii.** The empirical formula of the compound

12. (a) state a simple test that could be used to identify each gas in the laboratory
    i. Oxygen
    ii. Hydrogen
    iii. Carbondioxide
    (b) Two colourless gases at room temperature when mixing produce a white fume/smoke. Identify each gas and write the balanced chemical equation for the reaction

SECTION C (15 Marks)

Answer One question form this section.

13. Explain methods that are used to manage loss of plant nutrients form the soil (four methods)
14. In a certain chemical plant that deals with the production of hydrogen gas, a certain chemist passes an electric current in series through copper (II) sulphate and sulphuric acid solutions. In both electrolytes, inert electrodes were used. If $8400cm^3$ of hydrogen gas at STP were produced for several house, what volume of oxygen gas would also be produced? What mass of copper was produced?

QUESTIONS

THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM FOUR EXAMINATION - SERIES 08

CHEMISTRY

Time 3:00 Hrs **YEAR: 2022**

Instructions

1. This paper consists of sections A, B and C with the total of fourteen (14) questions.
2. Answer all questions in section A, B and one question from section C.
3. Cellular phones and any unauthorized materials are not allowed in the examination room.
4. Write your examination number on every page of your answer booklet(s)
5. The following constants may be used
6. **Remove this page and attach to the answer sheet provided**

Atomic masses= H = 1, O=16, S=32, Cl=35.5, Na=23, C=12, Ag=108.

Avogadro's constant:6.02×10^{23}

GMV at s.t.p =$22.4dm^3$

1 faraday=96,500C

1litre=$1dm^3$ =$1000cm^3$

SECTION A (15 marks)

Answer all questions

1. For each of the items i-ix, choose the correct answer from among the given alternatives and write its letter in the answer sheet provided.

   i. Loose or floppy clothing is not allowed in the laboratory. Why?

   A. Movement has to be fast
   B. It will get wet when water splashes
   C. It may catch fire or cause one to fall
   D. It causes poor ventilation in the body
   E. It prevents experiment from being conducted well

   ii. What mass of sulphuric acid (H_2SO_4) is found in 400 cm^3 of its 0.1M aqueous solution?

   A. 2.67g
   B. 9.8g
   C. 4.89g
   D. 3.92g
   E. 8.69g

iii. One of the isotopes of an element X has atomic number Z and a mass number A. what is the number of neutrons contained in the nucleus of the element?

A. Z - A
B. A
C. A + Z
D. Z
E. A – Z

iv. What is the proper set of apparatii would you use to measure various substances in the laboratory?

A. Stopwatch, beaker, burette and pipette
B. Beaker, pipette, stopwatch and electrical balance
C. Electrical balance, pipette, stopwatch and thermometer
D. Thistle funnel, pipette, burette and measuring cylinder
E. Beaker, measuring cylinder, thermometer and pipette

v. When methane undergoes substitution reaction with excess chlorine, what is the final product?

A. Chloromethane
B. Dichloromethane
C. Trichloromethane
D. Tetrachloromethane
E. Chloroethane

vi. Aluminum does not react with water and does not corrode much in air. Why?

A. It is below hydrogen in reactivity series
B. It forms a stable carbonate which prevent reactions
C. The metal is covered with protective coating of an oxide
D. It is very stable
E. Does not react with water

vii. Which of the following compound does not belong to the alkene homologous series?

A. C_2H_4 E. C_3H_6
B. C_4H_8
C. C_8H_{16}
D. C_5H_{12}

viii. Substance X liberates chlorine gas from acidified potassium chloride. What is the behavior described by X

A. Catalysis
B. An oxidation
C. Reduction
D. Decolourization

ix. When a burning fuel produces blue colour it means there is

A. Adequate supply of oxygen with production of soot.
B. Inadequate supply of oxygen without production of soot.
C. Inadequate supply of oxygen with production of soot.
D. Adequate supply of oxygen with production of more heat
E. Inadequate supply of oxygen with production of more heat

x. What is the number of molecules in 16.8dm^3 of chlorine at S.T.P?

A. $4.52X10^{23}$
B. $6.02X10^{23}$
C. $7.5X10^{23}$
D. $3.14X10^{23}$
E. $9.03X10^{23}$

2. Match the items in list A with the correct response in list B writing the letter of the correct response beside the item number in the answer sheet provided.

LIST A		LIST B
I.	A black solid element which burns with reddish glow giving off colourless gas which is slightly acidic.	A. Calcium B. Nitrogen
II.	Silvery white metal which burns with golden yellow flame giving an oxide which is basic in nature.	C. Aluminium D. Iron
III.	A yellow element in colour which burns with blue flame giving colourless gas which is strongly acidic in nature.	E. Carbon F. Sodium
IV.	A shinning white strip metal which burns with dazzling white flame giving an oxide which is slightly basic in nature.	G. Copper H. Beryllium I. Argon
V.	A slivery white metal which burns with brick red flame giving off oxide which is white solid.	J. Magnesium K. Sulphur
		L. Phosphorus M. Chlorine N. Zinc O. Fluorine P. Lead

3. a) Consider the following elements:
$^{16}O_8$, $^{19}F_9$, $^{20}Ne_{10}$, $^{23}Na_{11}$ and $^{24}Mg_{12}$. Atoms and ions of this elements can be Iso electronic (have the same number of electrons)

i. Write down their symbols when their Iso -electronic
ii. Write down their common electronic arrangement in their ions and atoms.

b) Although Sulphur dioxide is an oxide, it can be further oxidized

i. Write an equation and conditions for further oxidation of Sulphur dioxide
ii. Write down an equation showing how the product can be used industrially to obtain a desired material to be produced.
iii. Give the name for this industrial process
iv. Write down two uses of the product formed by the industrial process named in (iii) above.

4. Mwaisa guided students of form four to study carefully an experiment carried out in order to distinguish various substances formed during an experiment. The procedures were as follows

a) Measure about 120 cm^3 of water and pour it in a beaker
b) Add a spatulaful of common salts and stir to allow all salts to dissolve.
c) Continue to add more salts to the solution, stirring until no more salts can dissolve.
d) Place the solution on tripod stand and heat gently, as you continue stirring.
e) Stop heating when salts dissolves.
f) Place the beaker in a trough of cold water and allow cooling for 3 to 5 minutes. Note the result observed.

Questions

i. What type of solution was obtained when a spatulaful of table salt were dissolved in 120 cm^3 of water at room temperature? Give reason
ii. What type of solution was obtained when no more salts dissolved in cold water at room temperature? Give reason
iii. What is the name of the final solution obtained after cooling in trough of cold water? Give reason
iv. Explain the two (2) applications of the concept studied above

5. a) Give three application of separation of mixture to daily life.
   b) 20.0cm^3 of sodium hydroxide containing 8.0gdm^{-3} was required for complete neutralization of 0.18g of dibasic acid. Calculate the relative molecular mass of the acid.
6. a) State Faraday's laws of electrolysis
   b) Dilute Silver nitrates solution was decomposed by the passage of electric current through it. What mass of Silver and what volume of Oxygen (measured at (s.t.p) would be liberated in electrolysis by 9650 Coulombs of electricity?
7. Sulphuric acid act as dehydrating agent, oxidizing agent as well as an acid. Decide by a very brief answer how it behaving in each of the following cases.
   i. It evolves hydrogen gas when reacting with magnesium.
   ii. It is used to dry hydrogen chloride gas.
   iii. It reacts with sodium hydroxide to form salt and water.
   iv. It releases Sulphur dioxide gas when reacting with copper.
   v. It turns sugar black
   vi. It turns copper sulphate white
   vii. It produces carbon dioxide when added to the carbonate.
8. a) What do you understand by the following terms
      i. Mole concept
      ii. Molar volume of gas
   b) Consider the equation below for dissociation of sulphuric acid

      $H_2SO_4(aq) \longrightarrow 2H^+(aq) + SO_4^{2-}(aq)$

      From the equation how many ions are there in 9.8g of sulphuric acid?
9. a) Why petrol tanks in petrol station are constructed underground?
   b) What would happen to a well stoppered bottle full of water left in a deep freezer overnight? give reason to your answer
   c) Why iron not usually used in steam pipes and boilers?
   d) Explain by giving reasons why
      i. Laboratory door opens outward

ii. Laboratory floor is rough and never polished
iii. Fume chamber is important in the chemistry laboratory
iv. Some chemicals/ reagents are stored in brown coloured bottles

10. a) Write the chemical formula of the following compound
    i. Magnesium hydrogen carbonate
    ii. Sodium carbonate
    iii. Silicon (iv) oxide
    iv. Calcium nitrate

    b) Give the reason for your answer. What will happen when
    i. Barium chloride solution is added to an aqueous solution of potassium Sulphate
    ii. Silver nitrate solution is added to aqueous solution of potassium chloride
    iii. Concentrated - sulphuric acid is exposed to the atmosphere

11. a) In which other areas do we find the warning signs out of laboratory. Give four points.

    a) It is recommended that laboratory apparatus should be properly washed or wiped after use. Explain the significance for this when:

    b) i. Measuring volume of liquid
    ii. Measuring mass of substance

    c) With examples give two criteria used in classification of fuels.

12. a) Briefly explain the role if the following substance in the extraction of iron.
    (i) Coke
    (ii) Limestone

b) Solid iodine and gaseous hydrogen were kept in a closed system and heated until the following equilibrium was established.

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$.

If the water was introduced in the system hydrogen iodide dissolve in it.

i. In which direction would the equilibrium position be shifted?
ii. What would happen to the production of hydrogen iodide upon the introduction of water in the reaction system.

SECTION C (15 MARKS)

Answer only one question from this section

13. (a) Describe the extraction of sodium from its ore and write all the reaction equation
    (b) State four uses of sodium metal

14. A local industrialist came to your school to search information on the following concepts
    i. Land pollution
    ii. Eutrophication

    - Explain to her, what these concepts means
    - What are the causes, effects and precautions she is supposed to take to avoid their effects.

QUESTIONS

THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM FOUR EXAMINATION - SERIES 09

CHEMISTRY

Time 3:00 Hrs **YEAR: 2022**

INSTRUCTIONS:

1. This paper consists of section A, B and C with a total of fourteen (14) questions
2. Answer all questions in section A and B and one (1) question from section C
3. Cellular phones and any unauthorized materials are not required in the examination room
4. Non programmable calculators may be used
5. Write your examination number on every page of your answer sheet(s)
6. The following constants may be used

Atomic mass: H = 1, Na = 23, O= 16, K= 39, Mg= 24, Cl= 33.5, S= 32, C= 12, N=

- Cu= 63.5
- Avogadro's number = 6.02×10^{23}
- GMV at S.T. P= 22.4 dm^3
- Standard pressure = 760mmHg
- 1faraday = 96,500coloumbs
- $1dm^3 = 1litre = 1000cm^3$

SECTION A (15 MARKS)

Answer all question in this section

1. **For each of the item (i) –(x) choose the correct answer from among the given alternatives and write its letter beside the item number in the answer sheet provided**

(i) P and Q are atoms of elements in the same period. P is in group I and Q is in group II which of the following statements is **not** true about P and Q

A. P and Q have the same number of energy levels.
B. Q has one more proton than P in its nucleus
C. P has one more electron than Q in its valency shell
D. The atomic number of Q is greater than that of P
E. P and Q have equal number of neutrons

(ii) Which of the following pairs of oxides are gaseous at room temperature?

A. Carbon dioxide and copper (ii) oxide
B. Sulphur dioxide and copper (ii) oxide
C. Copper (ii) oxide and lead (ii) oxide
D. Sulphur dioxide and carbon dioxide
E. Iron (iii) oxide and carbon dioxide

(iii) In electrolysis, the equivalent of an element is the mass of the element liberated by:

A. 1 coloumb of electricity
B. 96,500 coloumbs of electricity
C. 2 coloumb of electricity
D. 9650 coloumb of electricity
E. Electrolytic process

(iv) What is the name of the alkene shown below?

$CH_3\ CHCH_2C = CH_2$
$\quad\ C_2H_5$

A. 2- methylpent – 4 – ene
B. 2-methylpentene
C. 4- methylhex – 1-ene
D. 4- ethylpent – 1-ene
E. 4 -ethylpent- 4-ene

(v) Which of the following is not true about bases?

A. Most bases have bitter taste
B. Bases are slippery when rubbed between fingers
C. Bases neutralize acids to form salt and water only
D. Weak base is the one that dissociate completely in solution
E. Are compounds which have oxide or hydroxide

(vi) When a burning fuel produces colour it means there is

A. Adequate supply of oxygen with production of soot
B. Inadequate supply of oxygen without production of soot
C. Inadequate supply of oxygen with production of soot
D. Inadequate supply of oxygen with production of less heat
E. Adequate supply of oxygen with production of more heat

(vii) It is possible to collect oxygen by downward displacement of water, because oxygen is

A. Denser than water
B. Soluble in water
C. Slightly soluble in water
D. Floats in water
E. Insoluble in water

(viii) If two gas jars labeled A and B contains 22.4dm^3 of oxygen and 22.4dm^3 of nitrogen respectively at STP, then it is true that

A. There were 6.02 x 10^{23} molecules of oxygen in A and 6.02 x 10^{23}molucules of nitrogen in B
B. There were 6.02 x 10^{23} atoms of oxygen in A and 6.02 x 10^{23}atoms of nitrogen in B
C. There were 12.04 x 10^{23} molecules of oxygen and nitrogen in their respective jars
D. 6.02 x 10^{23} molecules of oxygen in A and 6.02 x 10^{23} molecules of nitrogen in jar A and B respectively
E. Jars A and B contains 96,500 molecules of oxygen and nitrogen respectively

(ix) One of the following chemical equation is an example of the precipitation reaction.

A. $AgNO_3$ (aq) + $NaCL_{(aq)}$ ⟶ AgCL(s) + $NaNO_3$ (aq)
B. NaOH (aq) + HCL (aq) ⟶ NaCL (aq) + H_2O (l)
C. $CaCO_3$(s) ⟶ CaO(s) + CO_2(g)
D. Zn(s) + 2HCL (aq) ⟶ $ZnCL_2$ (aq) + H_2 (g)
E. C_2H_6(g) + O_2(g) ⟶ 2CO_2 (g) + 3H_2O(l)

(x) Which of the following is an agricultural chemical product made by application of chemistry?

A. Drugs B. Pesticides C. Clothes D. Yeast E. Cement

2. Match the items in LIST A with the correct responses in LIST B, by writing the letter of the correct response beside the item number in the answer sheet provided

LIST A	LIST B
(i) The strongest oxidizing agent	A. Chlorine
(ii) The third oxidizing agent	B. Iodine
(iii) The weakest oxidizing agent	C. Astatine
(iv) The second oxidizing agent	D. Fluorine
(v) Decreasing oxidizing power	E. F >CL>Br>I>As
	F. Bromine
	G. Oxygen
	H. I>Br >CL>F

SECTION B (70 MARKS)
Answer all questions in this section

3. (a)State whether each change below is a physical change or chemical change

(i) Glass bottle breaking
(ii) Coal burning in air
(iii) Dissolving sugar in ethanol
(iv) Butter and sugar being made into toffee
(v) Wet hair drying

(b) Coal contains the impurity of sulphur. When sulphur burns in air it forms sulphur dioxide. What volume of this polluting gas is produced at S.T.P when 1.0gof sulphur burns? (7marks)

4. (a) Write down the names and formulae of three common acids in the laboratory
   (b) Write down the products formed when each of the following pair of compounds reacts
   (i) Acid and metal
   (ii) Acid and metal carbonate
   (iii) Acid and metal hydroxide

   (c)Describe a test for identifying the gaseous product in b(i) and b(ii)

5. (a) Hydrated sodium carbonate ($Na_2CO_3.YH_2O$) solution was made by dissolving 178g of its crystals to make a litre of solution. $20cm^3$ of the solution was found to neutralize $25cm^3$ of a molar solution of nitric acid solution. Show all procedures to find the value of Y in the formula of hydrated carbonate

   (b) Name the indicator you would choose for the following titration
   (i) Hydrochloric acid versus ammonia solution
   (ii) Acetic acid versus sodium hydroxide

6. (a) Define the following term
   (i) Empirical formula
   (ii) Molecular formula

   (b) Chemical analysis show that the empirical formula of the compound is CH_2O and its relative molar mass is 60
   (i) Calculate its molecular formula
   (ii) Name the compound formed and write its open structural formula

   (c) Write balanced chemical equation of the reaction between the compound named in 6 (b) (ii) above with
   (i) Ethanol (C_2H_5OH)
   (ii) Sodium hydroxide

7. (a) In our daily life activities we use different types of fuels for various purposes. What is the energy value of a fuel?

   (b) Name two (2) example of each of the following types of fuels
   (i) Solid fuel
   (ii) Gaseous fuel
   (iii) Liquid fuel

(c) (i) Why coal is not recommended as domestic fuel?
(ii) List down advantage of liquid fuel over solid fuels (2 advantages)
(iii) What are two (2) gaseous fuel used in industries?

8. (a) Distinguish between the following
(i) Endothermic reaction from exothermic reaction
(ii) Reversible reaction from irreversible reaction

(b) The equation at which ammonia is manufactured is
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH3(g) \quad \Delta H = -46.2KJ/mol.$
The equilibrium concentration of ammonia increase as the temperature is lowered
(i) Is heat evolved or absorbed when ammonia is formed?
(ii) What will be the effect on the equilibrium concentration of ammonia if temperature and pressure are increased?

(c)List factors which can alter the rate of a chemical reaction (four factors)

9. (a) Give the name of the metal which meet the following information
(i) The metal which must be kept in kerosene to protect it from air and water
(ii) The metal which is found in limestone
(iii)The metal in green coloured which gives a black residue on heating or black oxide
(iv)The metal of which its oxide is yellow when hot and white when cold

(b) The following chemical substances were found in a school laboratory of Nindi secondary school, which warning sign is likely to be found on each of the following containers:
(i) Mosquito spray
(ii) Concentrated sulphuric acid
(iii) A bottle of lead nitrate.

10. (a) **Copy and complete the following table. It shows the products of electrolysis of aqueous solutions using carbon (inert) electrode**.

Solution	**Product at anode**	**Product at cathode**
Copper (ii) sulphate		
Sodium chloride		
Dilute sulphuric acid		

(b) What current must be supplied in a refinery to produce one metric tone (1000kg) of copper electrolytically within 24 hours?

(c) Suggest one method of separating the following mixtures
(i) Green solution from leaves
(ii) Ammonium chloride from sand
(iii) Ethanol and distilled water
(iv) Cooking oil and drinking water

11. (a) For each of the following plants, which nutrient do you think is missing?
(i) The plant has poor root growth
(ii) Plant does not flower, parts of the leaves are dead
(iii) The plant has very pale-yellow leaves
(iv) The plant has yellow leaves and very poor growth

(b) Distinguish manure from fertilizer.

(c) With the aid of a chemical equation, briefly explain how
i. temporary hardness of water can be removed by boiling
ii. permanent hardness of water can be removed by chemical means

12. (a) (ii) Briefly explain four causes that contribute to water pollution.
(b) Give any three point of Dalton atomic theory.

SECTION C (15 MARKS)
Answer only one question from this section

13. (a) Describe the extraction of sodium from its ore and write all the reaction equation
(b) State four uses of sodium metal

14. (a) When a white powdered sample A was strongly heated, a residue B which was yellow when hot and white on cooling was formed, and Gas C which formed white suspension F when reacted with calcium hydroxide solution. The residue B dissolved in dilute HCl to form solution D, when aqueous sodium hydroxide solution was added dropwise to solution D white ppt E soluble in excess alkali was formed
(i) Identify substance A, B, C, D, E, F and give their chemical formulae
(ii) Write balanced chemical reaction for the formation of gas C and precipitate E

(b) Give three uses of precipitation reaction

(c) Silver chloride is used to coat photographic paper
(i) What solutions could you mix to precipitate it?
(ii) Write a balanced equation for the precipitation of silver chloride

QUESTIONS

THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM FOUR EXAMINATION - SERIES 10

CHEMISTRY

Time 3:00 Hrs **YEAR: 2022**

INSTRUCTIONS

1. This paper consists of three sections A, B and C with total of **fourteen (14)** questions.
2. Answer **all** questions from section A and B and **one (1)** question from section C.
3. Calculators and cellular phones are **not** allowed in the examination room
4. Write you **Examination number** on every page of your answer sheets/booklet(s)
5. The following constants may be used:

   Atomic masses: H = 1, C = 12, N = 14, O = 16, Na = 23, S = 32, Cl = 35.5, Cu = 63.5 and Pb =207

   - Avogadro's number = $6.02 \times 10^{23} mol^{-1}$
   - GMV at s.t.p = 22.4 $dm^3 mol^{-1}$
   - 1 Faraday = 96500 Coulombs
   - Standard temperature = 273 K
   - 1 litre = 1 dm^3 = 1000 cm^3

SECTION A (15 Marks)

Answer **all** questions in this section

1. For each of the following items (i) – (x) choose the correct answer among the given alternatives and write its letter beside the item number in the answer sheet(s) provided

   (i) Splints are among the first aid kit components. Which of the following sentences best describes its use?

   A. Used for cutting bandages or piece of cloth
   B. Used for holding pieces of bandages or cloth together
   C. Used to support broken bones and is tied using bandages
   D. Used for picking up and holding things
   E. Used for cleaning wounds and make it heal faster

(ii) A simple proof that some chemical reactions take place in our bodies is that

A. We eat a balanced diet
B. Doctors tell us so in the hospitals
C. We occasionally fall sick
D. The food we eat or the drinks we take are quite different from the waste products from our bodies
E. There is no proof

(iii) The following reaction, $4NH_{3(g)} + 5O_{2(g)} \rightarrow 4NO_{(g)} + 6H_2O_{(g)}$, is an example of a

A. Decomposition reaction
B. Combination reaction
C. Redox reaction
D. Neutralization reaction
E. Esterification reaction

(iv) An aqueous solution turns red litmus solution blue. Excess addition of which of the following solution would reverse the change?

A. Baking powder
B. Lime
C. Aqueous ammonia
D. Vinegar
E. Quick lime

(v) Oils on treating with hydrogen in the presence of nickel catalyst form fats. This is an example of

A. Oxidation reaction
B. Addition reaction
C. Displacement reaction
D. Substitution reaction
E. Precipitation reaction

(vi) Three elements T, Q and R, are in the same period of the periodic table. The oxide of T is amphoteric, the oxide of Q is basic and the oxide of R is acidic. Which of the following shows the elements arranged in order of increasing atomic number?

A. T, Q, R
B. Q, T, R
C. Q, R, T
D. R, T, Q
E. T, R, Q

(vii) What happens when dilute sulphuric acid is poured on copper plate?

A. Zinc sulphate formed
B. Zinc chloride formed
C. Copper sulphate formed
D. Copper chloride formed
E. None of the above

(viii)The oxidation state of chlorine in potassium chlorate is

A. + 1
B. + 7
C. + 5
D. – 1
E. + 6

(ix)Ammonia gas is collected by which method among the following?

A. Downward displacement of water
B. Upward delivery
C. Downward delivery
D. Upward displacement of air
E. Collected over water

(x) Ten gram of copper powder were placed in a sealed initiation tube containing air. The tube was heated very strongly for half an hour. At the end of the experiment

A. Water vapour appeared in the tube
B. Undergoes decomposition
C. The total mass of the tube and its contents decreased
D. Copper melted into a liquid copper form
E. Copper increased in mass and changed in colour

2. Match the items in **List A** with the responses in **List B** by writing the letter of the correct response beside the item number in the answer sheet or booklet provided.

List A	List B
(i) When air is pumped into test tube contain lime water, lime water turns milky (ii) Condition for iron to rust (iii) Oxygen, Heat and Fuel (iv) Coating of Iron or steel with Zinc (v) Flammable liquids like petrol, alcohol and kerosene	A. Water is used for extinguishing B. Oxygen and water/moisture C. Presence of Oxygen D. Fire triangle E. Presence of carbon dioxide F. Rusting G. Galvanization H. Class B fire I. Nitrogen

SECTION B (70 Marks)

Answer **all** questions in this section

3. (a) The table below shows two brands of bottled water for drinking and the concentration of different mineral ions in each brand. Study the table and answer the questions below it.

Composition in mg/Litre	Mineral	Na^+	Ca^{2+}	Mg^{2+}	Cl^-	NO_3^-	SO_4^{2-}	Fe^{2+}	F^-
	Uhai	40.0	3.05	4.15	14.18	0.48	10.0	0	1.76
	Dasani	22.32	2.69	0.11	6.50	1.0	8.0	0	0.45

(i) Which brand of water is more hard? Explain.

(ii) State the benefit of having calcium ions in water;

(iii) Tap water is usually treated before being used. State what is added to perform that function.

(b) Hydrogen and phosphorus are non-metallic elements

(i) Which one between the two atoms is more electropositive?

(ii) Show your work clearly, write the chemical formula and the name of the compound formed when the two atoms combine.

4. One of the methods used for the preparation of oxygen is by the decomposition of hydrogen peroxide as shown in the reaction equation below

$2H_2O_{2(l)} \xrightarrow{MnO_2} 2H_2O_{(l)} + O_{2(g)}$

(a) (i) What is the purpose of MnO_2 in the reaction?

(ii) What will happen to the rate of the reaction if MnO_2 is not used in the above reaction?

(b) Briefly explain how the gas produced from the chemical reaction above is tested

5. (a) State the main raw materials and process involved in the manufacture of each of the following products

(i) Wood charcoal

(ii) Coke

(iii)Lamp black

(iv)Animal charcoal

(b) Name two main products formed when

(i) Hot concentrated sulphuric acid is added to sugar cane

(ii) Hot concentrated sulphuric acid is added to ethanol

6. (a) Define the following terms

(i) Mole

(ii) Molar mass

(b)112 dm^3 of oxygen gas was collected at S.T.P when a sample of lead nitrate was completely decomposed by heat. Calculate the volume of nitrogen dioxide produced.

7. A gaseous compound consists of 86% carbon and 14% hydrogen by mass. At s.t.p., 3.2 dm^3 of the compound had mass of 6 g. Calculate its molecular formula

8. (a) Study the following reaction equation

$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)} \quad \Delta H = -46.2\ kJmol^{-1}$

Use Le-Chatelier's principle, suggest how you would use temperature and pressure to obtain the highest production of ammonia at equilibrium

(b) The formation of methanol from hydrogen and carbon monoxide can be represented by

$CO_{(g)} + 2H_{2(g)} \rightleftharpoons CH_3OH \qquad \Delta H = 91\ kJmol^{-1}$

What mass of hydrogen would react to cause a heat change of 91 kJ

9. (a) Give the meaning of the following terms

(i) Soil pH

(ii) Liming

(b) (i) Explain why sulphur and its compounds are removed from the fuel before they are burned

(ii) By using a reaction equation explain how propane differs from propene

10. (a) Outline any three uses of carbon dioxide gas in animals and plants

(b) Complete the following reactions

(i) $CH_2=CH_2 + O_2 \longrightarrow ?$

(ii) $CaC_2 + 2H_2O \longrightarrow ?$

(iii) $CH_4 + Cl_2 \longrightarrow ?$

(iv) $C_2H_5OH + O_2 \longrightarrow ?$

11. (a) (i) What is the aim of having fume chamber in the chemistry laboratory?

(ii) Why laboratory doors are open outward?

(b) State the use of the following

Item		**Use**
(i)	Cotton wool	
(ii)	Petroleum jelly	
(iii)	Pain killers	
(iv)	Bandages	
(v)	Razor blade	

12. (a) With examples list four types of oxides

(b) Give any three types of salts

SECTION C (15 Marks)

Answer **only one** question in this section

13. Explain six (6) methods of preventing water pollution

14. In certain chemical plant that deals with the production of hydrogen gas, a certain chemist passes an electric current in series through copper (II) sulphate and sulphuric acid solutions. In both electrolytes, inert electrodes were used. If 8400 cm^3 of hydrogen gas at S.T.P were produced for several hours, what volume of oxygen gas would also be produced? What mass of copper was produced?